LIVE · SIMULATE · PREDICT

2026 · IPL · T20 · RV UNIVERSITY IOT PROJECT
ARDUINO UNO Q / ESP32 / BATTERY POWERED

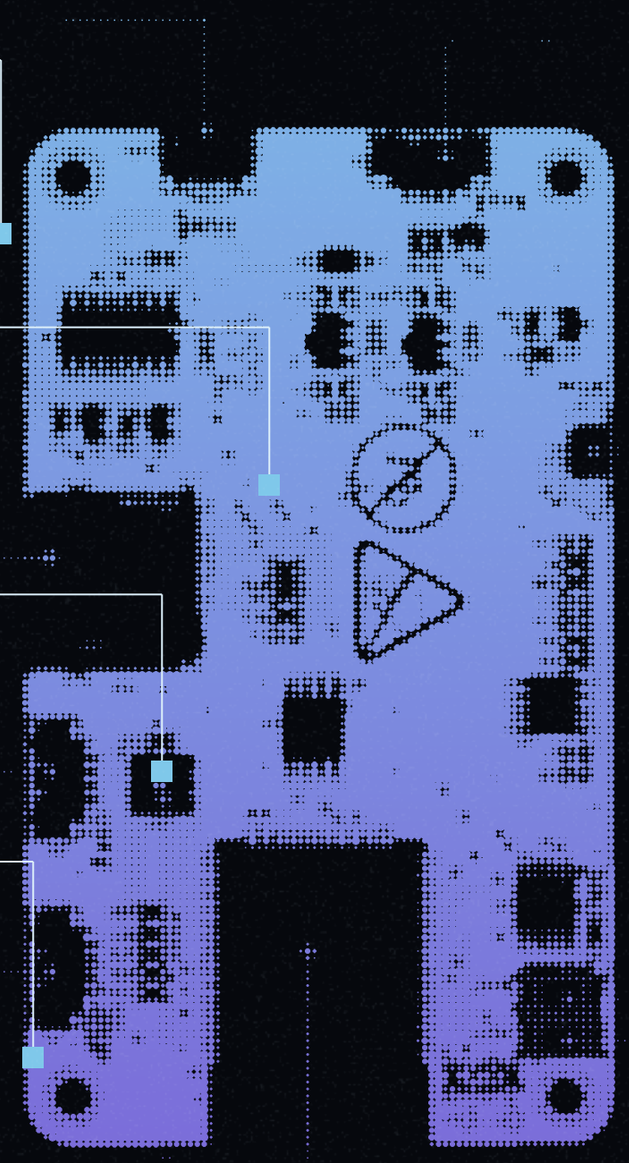
KHUSHI KRISHNA · HARSHUL KALA · MEGHNA PANT · MEDHA UMESH

ESP32-WROOM-32
the Cheap Yellow Display

2.8in ILI9341 · 320×240
XPT2046 touch

tap → live / simulation
hold → QR to the dashboard

P3 UART IO35 ← D1
IO22 → D0 · GND



unified cricket intelligence system

A battery-powered box that follows live IPL matches, plays new ones ball by ball and shows who is likely to win after every delivery. All of it runs on one Arduino UNO Q: a 2.8-inch touch screen shows the match at the table, and any phone on its Wi-Fi gets the full dashboard.

LIVE · SCREEN 1

the real match

The hub asks CricketData.org for the score every 150 s while a match is on and every 15 min otherwise, and stops two calls short of the free plan's 100 a day. With no IPL game on, Screen 1 shows the next fixture.

SIMULATE · SCREEN 2

a match nobody played

Pick two IPL franchises and the board bowls every delivery itself: dot 33.5%, four 11%, six 5.2%, wicket 5%, shifted by batter v bowler rating, the powerplay and the death overs. Each ball goes through the same scoring engine as a hand-scored one.

PREDICT · EVERY BALL

who wins from here

Runs still in reach with n balls and w wickets left: $E = Z(1 - e^{-1.6n/Z})$, where $Z = 20 \cdot w^{1.48}$. A logistic turns E against the runs needed into a win chance, held between 3% and 97%. The display (C++) and the hub (Python) run the same maths.

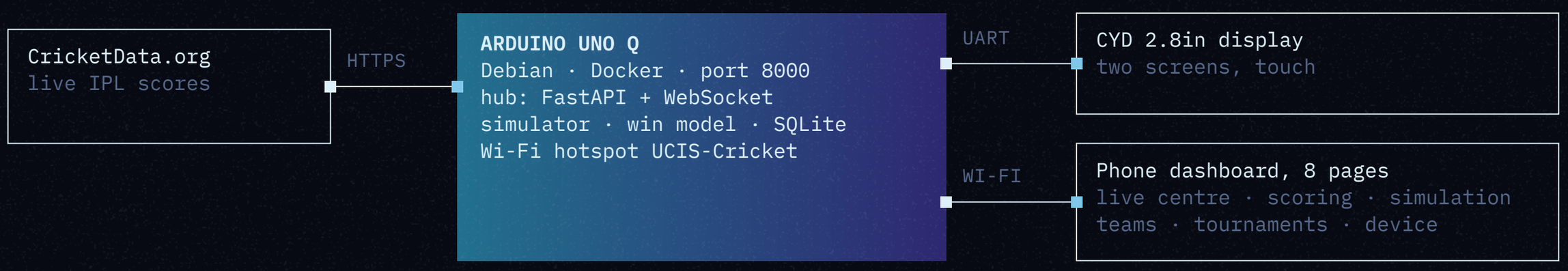
ONE SIMULATED MATCH · MI v CSK · SEED 25 · MI WIN CHANCE AFTER EVERY BALL



129/10
129/10

match tied.
match tied.
match tied.
match tied.

HOW IT IS WIRED



LAYER:01 — UNO Q MPU · DEBIAN · DOCKER
LAYER:02 — HUB · FASTAPI + WEBSOCKET
LAYER:03 — SIM · BALL-BY-BALL ENGINE
LAYER:04 — WIN MODEL · Z = 20·W^1.48
LAYER:05 — LINK · UART 115200 8N1
LAYER:06 — CYD · ESP32 + ILI9341
LAYER:07 — HOTSPOT · UCIS-CRICKET
LAYER:08 — DASHBOARD · REACT, PHONE

Join Wi-Fi **UCIS-Cricket** and scan the code, or open **10.42.0.1:8000**.

Next: player rating cards, a what-if replay, a crowd prediction game, LED flashes on fours, sixes and wickets, an Android app and a wireless display link.

ONE BAR PER BALL